Analysis & Prediction of Commodity Prices in Zambia
A Machine Learning Approach [James Mithiga: 58200]

The CRISP DM Methodology

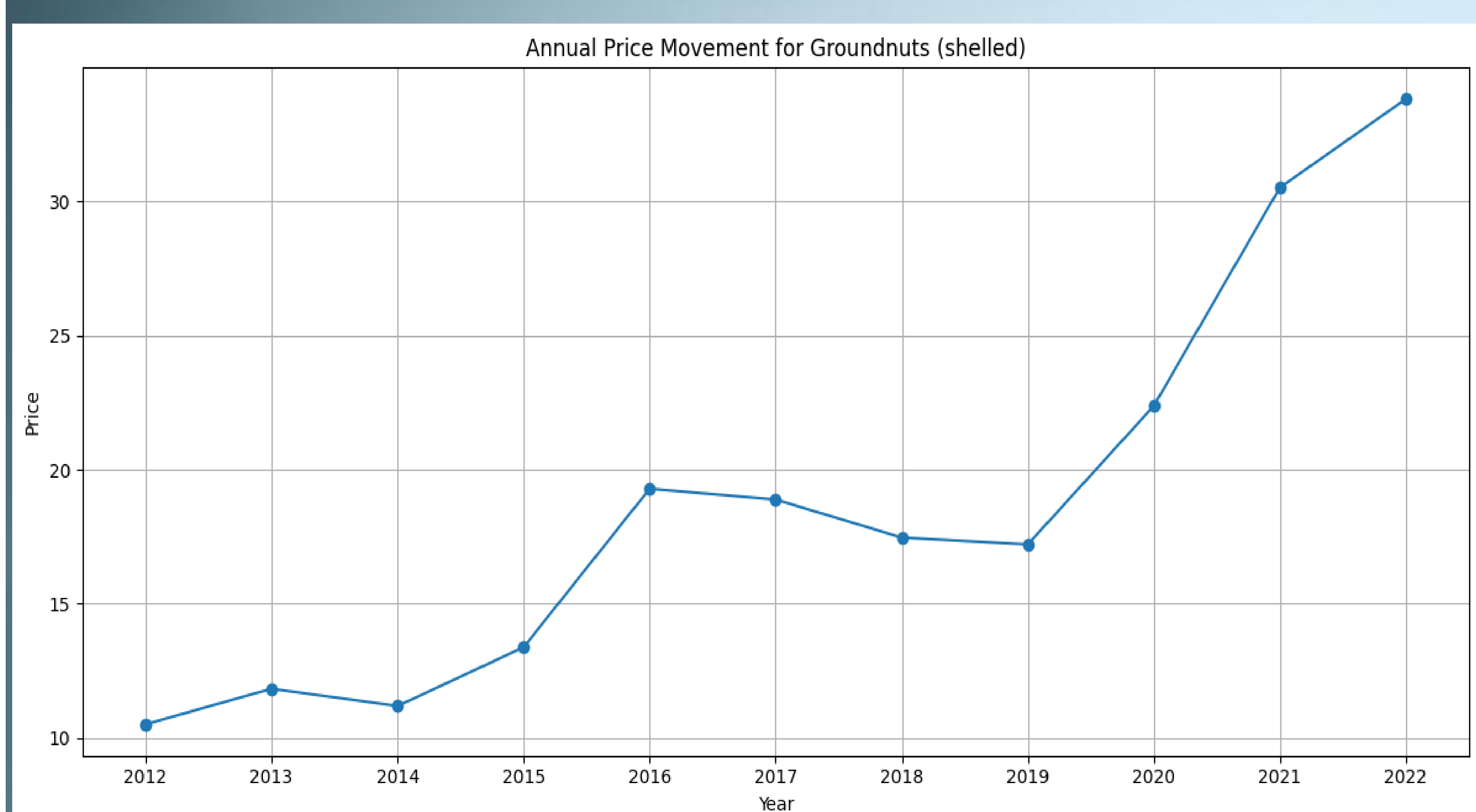
Business Understanding: The goal of the analysis is to understand and predict commodity prices over time.

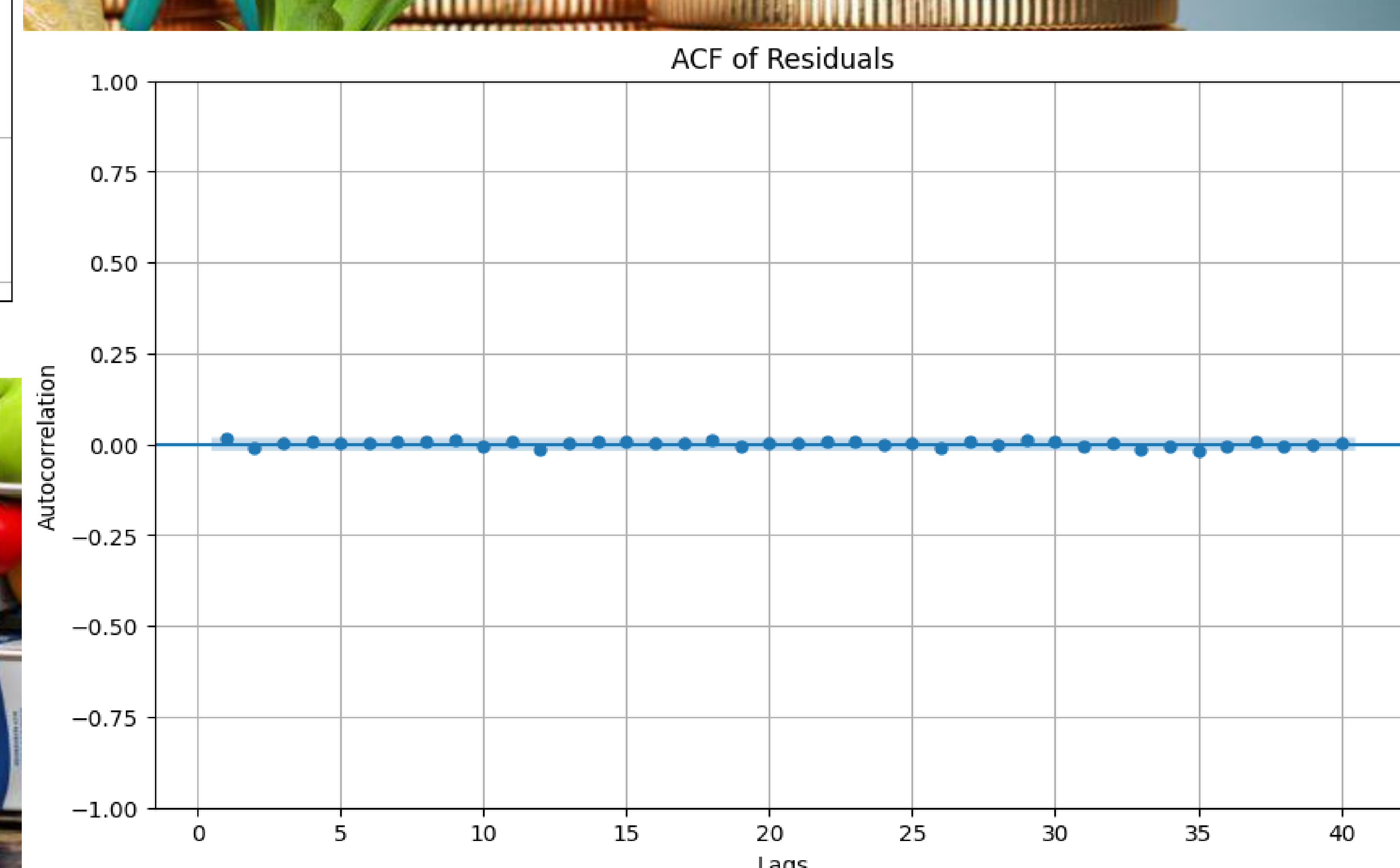
Data Understanding: Dataset contains attributes such as location, commodity category/name, price, month & year

Data Preparation: Several steps are taken to pre-process the data for modeling including feature engineering.

Modeling: A Random Forest Regressor is used to build the predictive model by Splitting the data, Training the model & making predictions of the commodity prices

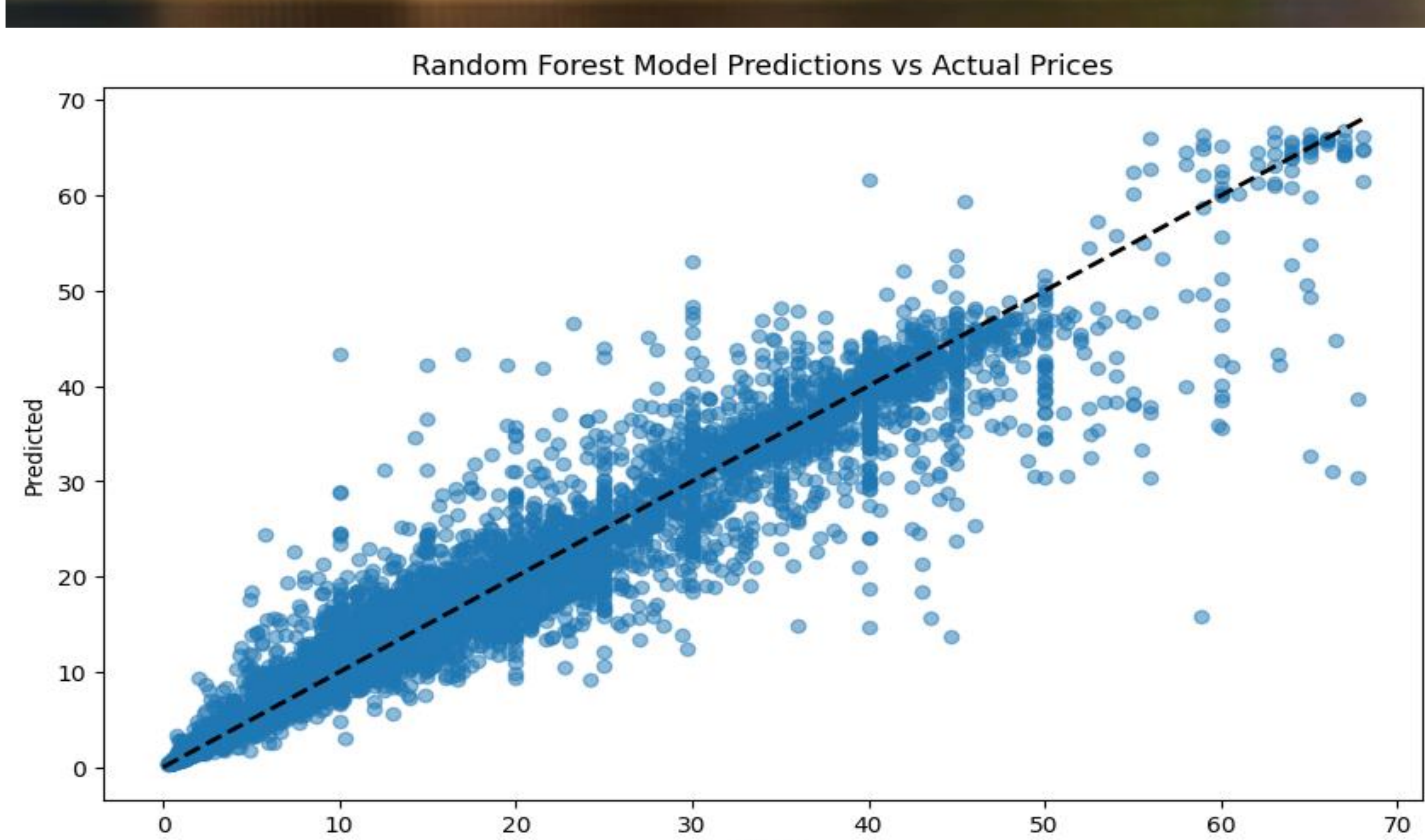
Link to Colab Notebook





Sample annual price movement of Groundnuts

ACF of Model Residuals



Prediction Model Evaluation

Features Selected:

- Geo Location (Latitude; Longitude)
- Commodity Category
- Commodity Name
- Month & Year

Evaluation Results

- Mean Squared Error: 9.54
- R-squared Score (Accuracy): 94%
- Root Mean Squared Error (RMSE): 3.09
- Mean Absolute % Error (MAPE): 11.79